Status of Digital Health Technology Adoption in Five Vietnamese hospitals: Cross-sectional Assessment

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Table of Contents

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Abstract

Background: Digital health technologies (DHTs) have been recognized as a key solution to help countries, especially those at the low- and middle-income group, to achieve the Sustainable Development Goals and the World Health Organization's Triple Billion Targets. In hospital settings, DHTs need to be designed and implemented taking into account the local context to achieve usability and sustainability. As projects such as the Vietnam ICU Translational Applications Laboratory are seeking to integrate new digital technologies in the Vietnamese critical care settings, it is important to understand the current status of DHT implementation in Vietnamese hospitals.

Objective: We aimed to explore the current digital maturity in Vietnamese public hospitals to understand their readiness in implementing new DHTs. An overview of DHT adoption in five Vietnamese hospitals were conducted, using the Vietnam Health Information Technology (HIT) Maturity Model.

Methods: We assessed the adoption of some key DHTs and infrastructure in five top-tier public hospitals in Vietnam, using a questionnaire adapted from the Vietnam HIT Maturity Model. The questionnaires were answered by the heads of the hospitals' Information Technology departments with follow-up for clarifications and verifications on some answers. Descriptive statistics demonstrated on radar graphs and tile graphs were used to visualize the data collected.

Results: Hospital information systems (HIS), laboratory information systems (LIS), and radiology information systems - picture archiving and communication systems (RIS-PACS) were implemented in all the five hospitals despite with varied maturity across the hospitals. Two hospitals were not using any electronic medical record system nor fulfilling any extra digital capability such as implementing clinical data repositories and clinical decision support systems (CDSSs). No hospital reported sharing clinical data with other organizations using Health Level Seven (HL7) standards such as Continuity of Care Document and

Clinical Document Architecture although two reported their systems adopted these standards. Almost all hospitals reported their RIS-PACS adopted Digital Imaging and Communications in Medicine (DICOM) standard.

Conclusions: The major public hospitals attended in this assessment showed relatively high levels of digital maturity that aligns with national frameworks and policies. It is promising that these hospitals will be able to implement artificial intelligence -enabled CDSSs that leverage a wide range of data generated from systems such as HIS, LIS, RIS-PACS and, in some cases, EMRs. The widespread adoption of international standards such as HL7 and DICOM suggests DHTs developed in Vietnam could be implemented in other countries.

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Abstract

Background: Digital health technologies (DHT) have been recognized as a key solution to help countries, especially those at the low- and middle-income group, to achieve the Sustainable Development Goals and the World Health Organization's Triple Billion Targets. In hospital settings, DHTs need to be designed and implemented taking into account the local context to achieve usability and sustainability. As projects such as the Vietnam ICU Translational Applications Laboratory are seeking to integrate new digital technologies in the Vietnamese critical care settings, it is important to understand the current status of DHT adoption in Vietnamese hospitals.
Objective: We aimed to explore the current digital maturity in 5 Vietnamese public hospitals to understand their readiness in implementing new DHT.
Methods: We assessed the adoption of some key DHT and infrastructure in 5 top-tier public hospitals in Vietnam, using a questionnaire adapted from the Vietnam Health Information Technology (HIT) Maturity Model. The questionnaires were answered by the heads of the hospitals' Information Technology departments with follow-up for clarifications and verifications on some answers. Descriptive statistics demonstrated on radar graphs and tile graphs were used to visualize the data collected.
Results: Hospital information systems (HIS), laboratory information systems (LIS), and radiology information systems - picture archiving and communication systems (RIS-PACS) were implemented in all 5 hospitals despite with varied maturity across the hospitals. At least 50% of the LIS criteria in the Vietnam HIT Maturity Model were satisfied by the hospitals in the assessment. However, this threshold was only met by only 80% and 60% of the hospitals when it comes to HIS and RIS-PACS, respectively. Two hospitals were not using any electronic medical record system nor fulfilling any extra digital capability such as implementing clinical data repositories (CDR) and clinical decision support systems (CDSS). No hospital reported sharing clinical data with other organizations using Health Level Seven (HL7) standards such as Continuity of Care Document (CCD) and Clinical Document Architecture (CDA) although 2 reported their systems adopted these standards. Four out of 5 hospitals reported their RIS-PACS adopted Digital Imaging and Communications in Medicine (DICOM) standard.
Conclusions: The 5 major Vietnamese public hospitals in this assessment have widely adopted

information systems such as HIS, LIS, and RIS-PACS to support administrative and clinical tasks. While the adoption of EMR systems was less common, their implementation revolved around data collection, management, and access to clinical data. Secondary use of clinical data for decision support through the implementation of CDR and CDSS was limited, posing a potential barrier for the integration of external DHTs into the existing systems. However, the wide adoption of international standards such as HL7 and DICOM is a facilitator for the adoption of new DHTs in these hospitals.

Keywords: electronic health records; electronic medical records; digital maturity; clinical decision support; digital infrastructure; Vietnam; health information technology; digital health technology; low and middle income countries.

Introduction

The Covid pandemic demonstrated the needs for countries around the world to implement the national digital health infrastructure needed to respond to health emergencies and to achieve the Sustainable Development Goals (SDGs)[1]. As part of this digital transformation effort, digital health technologies (DHTs) such as Electronic Medical Record (EMR) systems, Laboratory Information Systems (LISs) and Picture Archiving and Communication Systems (PACSs) have become widespread. In recent years, Vietnam has developed and begun implementation of an ambitious national digital health strategy that includes the deployment of hospital-based EMRs and an electronic health insurance claim system[2].

As the cost of hardware continues to drop and the range and capabilities of DHTs continues to expand[3], digitization of health systems is becoming increasingly feasible in low- and middle-income countries (LMICs). New technologies such as the Internet of Things, wearable devices, and artificial intelligence (AI) have begun to be adopted in healthcare settings and could be an important enabler for realizing the WHO's triple billion targets (more than one billion people benefiting from Universal Health Care; better protected from health emergencies; and enjoying better health and wellbeing) and to accelerate SDGs achievement[4]. This year, the WHO announced the Global Initiative for Digital Health, to help LMICs achieve the aims of the Global Strategy on Digital Health by developing their national digital health infrastructure, adopting international data standards and fostering increased international collaboration in digital health[1, 4].

In global health research, there is increasing interest in implementing new technologies such as AI or smart wearables for clinical decision support[5, 6]. However, for these systems to be practicable and to take advantage of increasingly available "big data", the local provision of technological infrastructure and implementation capabilities is necessary. For example, hospitals that have implemented EMR systems could use data from the EMR to train AI models and enable the implementation of clinical decision support systems (CDSSs)[7]. In addition, the adoption of international interoperability standards, such as Health Level Seven (HL7) Fast Healthcare Interoperability Resources (FHIR), is also necessary to allow efficient data exchange between information systems and new DHTs[8].

In hospital settings, digital maturity models such as the Electronic Medical Record Adoption Model (EMRAM) from the Healthcare Information and Management Systems Society (HIMSS) are often used to guide digital health technology adoption and implementation[9]. These models can be employed on a national or international scale and can target different dimensions including technology, strategy, interoperability, analytics, and governance. There is not currently an international consensus on how to best measure digital capabilities in healthcare institutions[9, 10] and variations in maturity models across different countries may reflect an adaptation to national priorities and contexts, especially when being implemented in LMICs. In 2017, the Vietnam Ministry of Health (MoH) issued the Vietnam Health Information Technology (HIT) Maturity Model, originally known as Circular 54/2017/TT-BYT, as a roadmap for Vietnamese public hospitals towards a "paperless" and "smart" hospital model[2, 11]. While this guidance defines 7 maturity stages similar to EMRAM, the criteria to achieve a specific maturity level were significantly modified. For example, the Vietnam model includes the requirement for hospital information system (HIS) to connect to the Vietnam Social Security claim portal and assesses the integration of the national standardized vocabularies into the digital systems[11].

The Vietnam ICU Translational Applications Laboratory (VITAL) project focuses on developing new technologies for improving outcomes of critically ill patients in Vietnam such as CDSS utilizing AI algorithms trained on local data[12-14]. Understanding the national DHT policy landscape including regulations, infrastructure, and professional workflows are important elements[2, 15] to ensure the sustainability and successful implementation of the technologies[6,16]. For this reason, we aimed to explore the current status and likely short-term development of DHT infrastructure in 5 major public hospitals in Vietnam. Whilst the findings may be unable to generalize about the state of DHT adoption in all Vietnamese hospitals, we expect they can inform the discussions around how new DHT can be integrated with the existing IT infrastructure in Vietnamese public hospitals.

Methods

This study utilized a structured questionnaire to explore the current adoption of DHT in five public hospitals in Vietnam. The assessment items were developed based on the Vietnam HIT Maturity Model.

The Vietnam HIT Maturity Model

The Vietnam HIT Maturity Model comprises of eight domains related to technology adoption and digital capabilities in hospitals[11]. The eight domains are known as IT Infrastructure, Hospital Information System (HIS), Laboratory Information System (LIS), Radiology Information System - Picture Archiving and Communication System (RIS-PACS), Electronic Medical Record (EMR), Administrative and Operation Software, Security and Information Safety, and Non-functional Criteria. Except for Security and Information Safety and Non-functional Criteria, each domain includes a set of infrastructure requirements such as having a server room, and/or software functional criteria such as having lab tests management functionality. The Model also sets out further digital capabilities focusing on assisting clinical practice such as CDSS, clinical data repositories (CDR), and medication management. These are named *Extra capabilities*.

Following a stage-based approach, a hospital would reach a maturity level in a domain if a predefined set of criteria is met. They can only move to a higher maturity level when all criteria of the lower levels have been satisfied. The domain *IT Infrastructure* and *Hospital Information System* have 7 maturity levels, from *Level 1* to *7*, while the other domains have 2 levels, *Basic* and *Advanced*. (Figure 1) presents the maturity levels and their scoring criteria based on the domains and extra capabilities.

Figure 1. The Vietnam Health Information Technology Maturity Model issued 2017 by the Vietnam Ministry of Health[11]. The header column shows the maturity levels while the header row shows the domains and extra capabilities that make up the maturity levels.

	IT infrastructure	HIS	LIS	RIS-PACS	EMR	Administration and Operation Software	Security and Information Safety	Non-functional Criteria	Extra capabilities
HIT Level 7	Level 7	Level 7	Advanced	Advanced	Advanced	Advanced	Advanced	Advanced	- "Paperless" hospital if all relevant criteria are met - CDSS level 3 supporting doctors' decisions related to treatment protocols and treatment results using suitably customized templates - Data in CDR is analysed to improve care quality, patient safety, and care efficiency - Clinical data can be readily shared for stakeholders in care coo rdination based on HL7 standards - Continuous reports of hospital services using the data collected
HIT Level 6 (Smart hospital)	Level 6	Level 6	Advanced	Advanced	Basic	Advanced	Advanced	Advanced	- CDSS level 2 providing: evidence-based warnings for treatment i.e. health and medicine advices, drug information/interaction check, and initial order/prescription violation indentification rules - All structured forms i.e. progress notes, consultation notes, problem lists, and discharge summaries are digitalized - Closed-loop management of drugs, using identification technologies to assist drug administration
HIT Level 5	Level 5	Level 5	Advanced	Advanced	n/a	Basic	Basic	Basic	PACS can replace physical films
HIT Level 4	Level 4	Level 4	Advanced	Basic	n/a	Basic	Basic	Basic	- PACS allows doctors to access images outside the imaging department - Electronic ordering - Electronic management of inpatient orders
HIT Level 3	Level 3	Level 3	Basic	n/a	n/a	Basic	Basic	Basic	- Electronic records having digital vital sign records, nursing notes, medical procedures, and surgical procedures are stored in CDR - CDSS level 1 assisting electronic prescription (new or historic prescription) - Pharmacy information available in the hospital network and supported with CDSS
HIT Level 2	Level 2	Level 2	n/a	n/a	n/a	n/a	n/a	n/a	- A CDR consisting the normenclature and coding systems, pharmacy, orders, and test results (if available) - Data in CDR can be shared between stakeholders for care coordination
HIT Level 1	Level 1	Level 1	n/a	n/a	n/a	n/a	n/a	n/a	Patient's information can be accessed electronically

Questionnaire, data collection, and analysis

Since the focus of this study was the adoption of digital health systems in hospitals, we developed an assessment tool based on the criteria from the IT infrastructure, HIS, LIS, RIS-PACS, EMR, and Extra Capabilities domains (see Multimedia Appendix 1). The Administrative and Operation Software, Security and Information Safety, and Non-functional Criteria domains were not included in this study as we considered these to be less relevant to the study's focus despite their importance to the overall digital maturity of a healthcare institution.

For each domain, we gathered information through the hospital IT departments to determine whether the hospitals had met the component criteria or whether they had planned to achieve the criteria in the next 3 to 5 years. Results of the assessments were discussed with the heads of the hospital IT departments to ensure accuracy. The data collection was carried out between July 2022 to March 2023. Information on the maturity of each hospital was entered into Excel for initial data cleaning then analyzed using R statistical software[16].

Our aim was to provide a descriptive summary of what infrastructure and functionalities, or criteria, were available in each hospital across the domains rather than calculating scores for their digital maturity. The proportions of criteria that each hospital met in each domain was calculated. A radar graph were built using the fmsb package in R[17] to depict the difference in the proportions of criteria met among the hospitals, by each domain. Radar plots were regarded as an efficient tool to compare between various groups on multiple variables as discussed by Saary[18]. In addition to the summary statistics, we further compare the technology adoption between the hospitals by delving into the individual domains. Tile graphs were created using the ggplot2 package[19] in R to illustrate the criteria that the hospitals had satisfied within each domain. These graphs allow for the visualization of individual data points (e.g. which criterion was met by a specific hospital) as opposed to visualizing the counts or proportions of hospitals that met a particular criterion as in the case of bar charts.

Study population

Until the beginning of 2024, it was estimated that there were nearly 1500 public and over 300 private hospitals in Vietnam[20]. Five hospitals were purposively selected for this study based on previous interest in DHT innovation and participation in digital health research activities. Our approach aimed to sample hospitals where DHT innovation was likely to occur soonest, rather than provide a comprehensive view of all government hospitals. Of the hospitals that participated in this assessment, four were in either Hanoi or Ho Chi Minh City, the two largest cities in Vietnam, and one was in a Mekong Delta (i.e., rural) province. All were top-tier teaching public hospitals with a capacity of over 500 beds. Two were tertiary hospitals that focus on a limited number of specialties, while three were general hospitals that provide care in a wide range of specialties. Most hospitals had between 8 and 12 IT staff per 1000 beds, except for one hospital that had an approximate ratio of over 50 per 1000 beds. Characteristics of the hospitals are shown in (Table 1). To ensure confidentiality, we withheld information that could be used to identify the hospitals.

Table 1. Characteristics of the hospitals participating in this assessment.

Hospital	Tier	Types	Number of beds	Teaching hospital	IT staff/beds ratio
A	1	General hospital	Over 500	Yes	50 : 1000
B	1	Tertiary hospital	Over 500	Yes	13 : 1000
C	1	Tertiary hospital	Over 500	Yes	11 : 1000
D	1	General hospital	Over 500	Yes	8 : 1000
E	1	General hospital	Over 500	Yes	12 : 1000

Ethical considerations and recruitment

This project was exempted from ethical review as it constituted a Service Review as defined by the Oxford Tropical Research Ethics Committee (OxTREC)[21, 22]. The selected hospitals were identified from a professional network that the study's principal investigators (LT and CP) were part of. The hospitals were invited to participate in the assessment through an invitation letter explaining the purpose and scope of the study. Upon the approval of the hospitals' authorities, the hospital IT directors were provided with an information sheet, the questionnaire, and an instruction document. Queries regarding the assessment, if any, were clarified by the first author (DMT).

The IT directors were informed in writing that they could refuse to provide any data as they wished and the data collected would be securely managed and only be used for the purpose of the study. Participating departments were able to review and agree the contents of the manuscript prior to submission. No personally identifiable data was collected in the research. The research data are managed per the Oxford University Clinical Research Unit's data management policy. Information that can be used to identify the hospitals such as names and the number of beds were de-identified to ensure their confidentiality. There was no compensation for the hospitals participating in this study.

Descriptions of information systems in Vietnamese public hospitals

HIS, LIS, RIS, PACS, and EMR are popular information systems in healthcare institutions worldwide. However, their functionalities and use cases may differ to some extent depending on the national context that they are implemented in. In this section, we provide a Vietnam-oriented description of these information systems to support the interpretation of the study's findings.

Hospital Information System

Also known as hospital management systems, HISs are usually prioritized by Vietnamese hospitals among other information systems. They are typically used to manage outpatient services as well as to manage administrative and billing data emerging from all wards and departments in a hospital. HIS can be interfaced with radiology information systems and LISs to send orders or receive imaging and lab results. A significant proportion of HIS in Vietnamese hospitals have been developed and provided by local vendors such as The Corporation for Financing Promoting Technology (known as FPT), Vietnam Posts and Telecommunications Group (known as VNPT), and Viettel[23].

Laboratory information system

An LIS allows clinical laboratories to track and utilize a wide range of data related to orders, specimens, results, and consumables. It reduces the turnaround time from ordering tests to collecting laboratory results through automatic processes such as integrating with an HIS to transfer orders to the laboratory and connecting with laboratory instruments to automatically receive results. Clinicians can use an LIS installed at their ward or an HIS interfaced with an LIS to access laboratory results when available. Many public hospitals in Vietnam chose LISs developed by local vendors such as LABCONN by LABSoft (deployed in over 200 laboratories) and LABMDSOFT by MDsoft.

Radiology Information System - Picture Archiving and Communication System

RIS and PACS are often used together in medical imaging departments. The RIS typically supports managing orders, patients, and imaging procedure information. An RIS can receive order information and send images and reports back to the HIS using HL7 messages. The RIS can also send orders to imaging machines and receive processed images from PACS. The PACS is comprised of 4 components including imaging instruments, a secure network to transfer the images, workstation computers that allow doctors to view the images, and a storage system that can archive the images. A PACS can thus facilitate image viewing, editing, storing, and sharing. With a PACS, hospitals can eliminate physical film usage and provide multi-site access to medical images through a web interface.

Electronic Medical Record

In Vietnam, EMR systems refer to the systems implemented in hospitals, especially in inpatient wards, to manage patient care information such as medical history, problems, orders, tests, and medications, similar to paper charts[24]. EMRs are defined differently from electronic health records (EHRs) in Vietnam, which are defined as health records updated throughout a person's lifetime and can cover a comprehensive range of health information such as allergies, vaccination, family history, and outpatient visits rather than specialized care information[25]. While the data collected to EMR systems are independently managed by hospitals, EHR data are synchronized from multiple sources such as hospital and primary care facility visits to a centralized platform managed by the MoH[26]. In the Vietnam HIT Maturity Model, EMR systems were assessed on 4 areas as follows.

- Clinical data management functions, such as medical history, clinical document, and test management.
- Administrative and demographics functions, including managing information of health staff

and patients, and managing integrations between the EMR system and other information systems in a hospital.

- Medical record storage capabilities, including storage duration meeting requirements from Healthcare Law, record synchronization, and medical record storage and restoring.
- Technical administration functions, including security, supervision, standardized terminology management, standard-based data exchange, EMR workflow management, and database back-up and recovery.

Extra capabilities

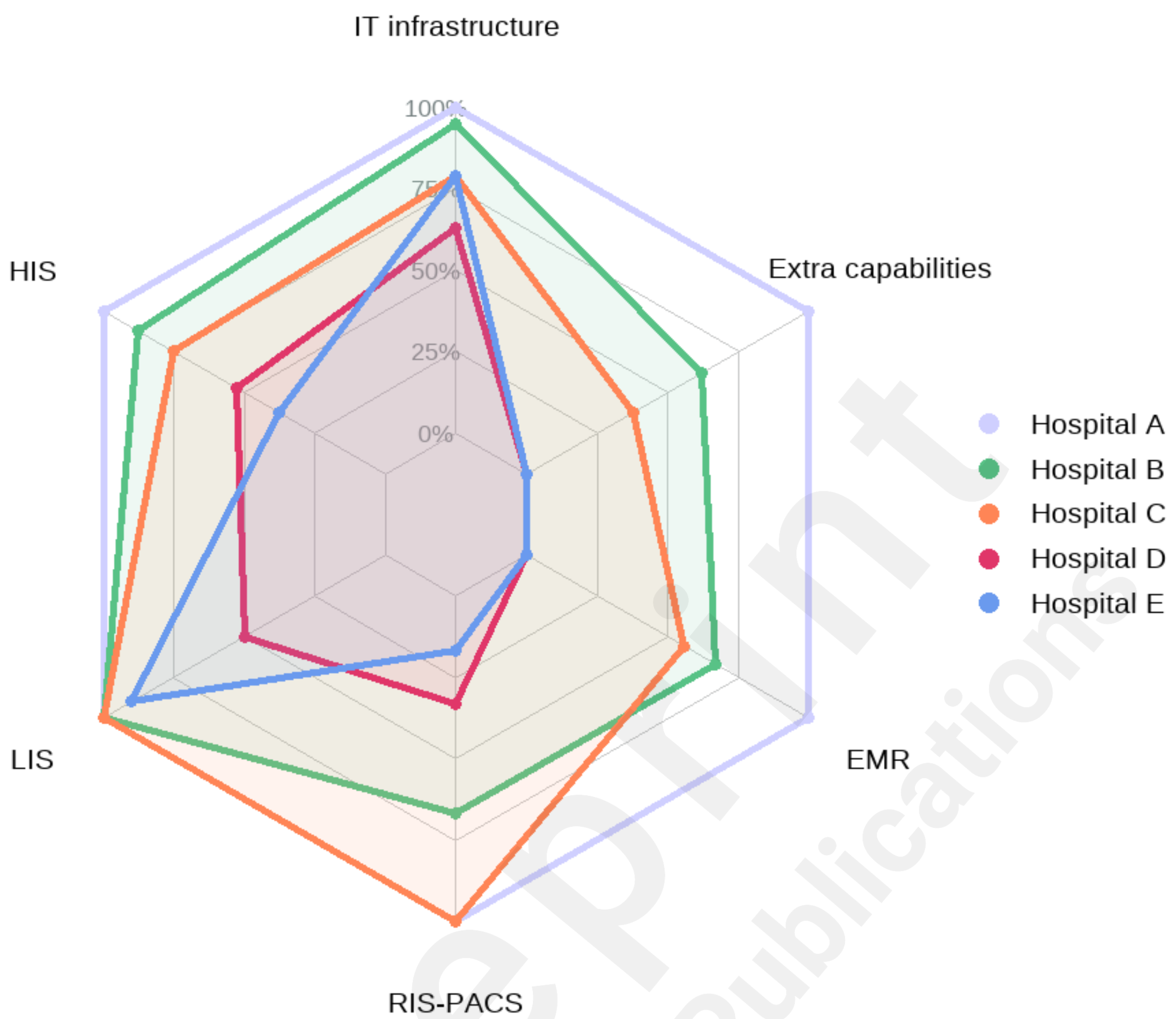
In the Vietnam HIT Maturity Model, each maturity level requires several extra digital capabilities in addition to the principal information systems such as the HIS, LIS, and EMR systems (see (Figure 1)). These capabilities can be grouped as follows:

- Clinical data repositories (CDR). These are centralized data stores that gather patient data generated from other clinical information systems. Data and resources from CDR can also be used by other systems and applications in the same network such as standardized lab order vocabularies can be jointly deployed in EMR and LIS. This assessment examined CDR implementation on the following criteria:
    - Hospitals can establish a CDR containing standardized vocabularies, medications, orders, and lab tests.
    - Information in CDR can be shared with stakeholders involved in patient care.
    - CDR contain data about vital signs, nursing notes, and clinical procedures.
- CDSSs are categorized into 3 levels:
    - CDSS level 1 can support drug prescribing including new prescription and re-prescribing.
    - CDSS level 2 can send alerts for basic conflicts in ordering and prescribing.
    - CDSS level 3 can inform clinicians’ treatment plans and outcomes via appropriately customized alerts.
- Electronic ordering and inpatient order management.
- Digitizing clinical notes using electronic structured templates.
- Closed loop medication management using identification technologies such as Radio Frequency Identification and barcoding.
- Sharing clinical data to stakeholders involved in patient care through standardized electronic transactions such as HL7 CDA and HL7 CCD.
- Continuous summaries of service utilization data from all the departments such as inpatient, outpatient, and emergency department.

Results

While we were not able to collect information about 3 to 5-year investment plans from the hospitals, we were able to collect data on all the assessment criteria from the maturity model. The radar plot in (Figure 2) presents the percentage of criteria the hospitals met in each domain.

Figure 2. The percentage of HIT criteria satisfied in each domain across the 5 hospitals (IT infrastructure, EMR, LIS, HIS, RIS-PACS, and extra capabilities).



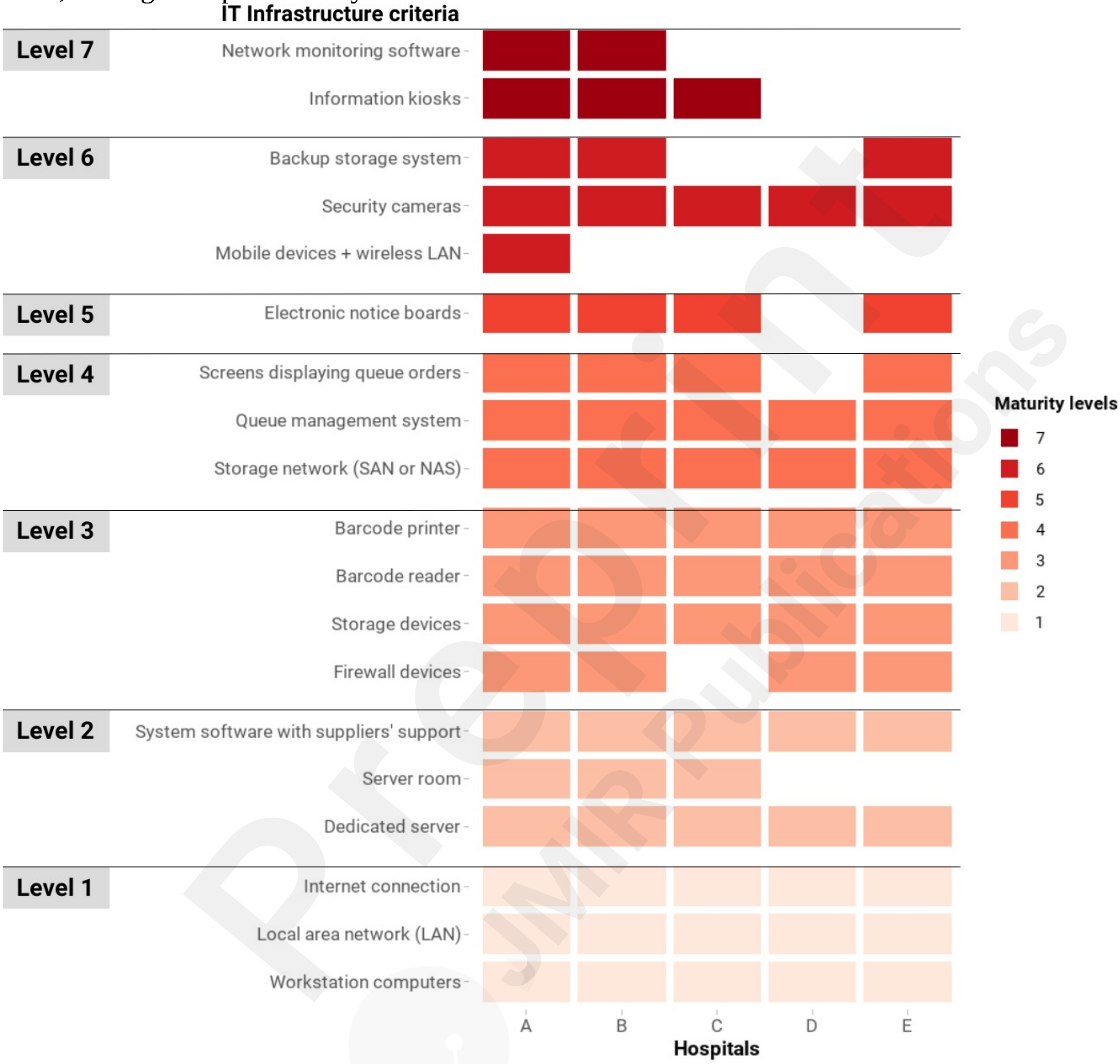
In general, HIS, LIS, and radiology information systems - picture archiving and communication systems (RIS-PACS) were implemented in all the hospitals despite with varied maturity. Two hospitals were not using any EMR system nor fulfilling any extra digital capability. One hospital reported met all the criteria across the 6 domains. In the following sections, we describe the domains in more detail and compare the implementation strategies between the hospitals.

IT Infrastructure

(Figure 3) shows the IT Infrastructure criteria that have been met by the five hospitals. Level 1 criteria were met by all hospitals and most hospitals reported meeting those for Level 2 to Level 5.
All the hospitals had sufficient workstation computers for staff. They were connected to the Internet and local area networks. All the hospitals' digital systems and databases were run on dedicated servers, and three hospitals had their own on-premises server rooms with physical controls. Hospitals' data were stored on storage devices that are supported by storage networks such as Network Attached Storage or Storage Area Network systems. Backup storage systems were available in three out of five hospitals.
Almost all hospitals in this assessment adopted technologies to help reduce errors and improve patient flows. These are barcode readers, barcode printers, queue management systems, screens displaying queue orders, and electronic notice boards. Information kiosks were less common as only 3 in 5 hospitals implemented this technology.
Level 6 and 7 criteria were less available in the hospitals. Only one hospital reported adopting mobile devices such as tablets and smartphones and a wireless local area network to support hospital

activities.

Figure 3. The criteria of IT infrastructure in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of level 7 can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.



Hospital Information System

All the hospitals reported their HIS fully met the Level 1 and Level 2 criteria (Figure 4). These systems were able to support patient registration, outpatient services, and pharmacy stock management. Hospital fees including social health insurance payments were also managed in these HIS, where insurance claims were formatted into XML files following a national standard and routinely submitted to the Vietnam Social Security portal. The HIS were reportedly able to integrate Vietnam's national coding systems to standardize the classifications of some administrative, such as the Vietnam ethnic classification system, and clinical services, such as the WHO International Classification of Disease[27]. Medical orders and test results were assigned unique patient IDs and gathered in HIS. Inpatient service management and reporting functionalities (Level 3) were also available in the HIS across 5 hospitals.

Four out of five hospitals reported that their HIS integrates with PACS and cashless payment technology. However, modules for treatment protocols, professional procedures, and drug interaction management were implemented in only three hospitals.

We found the functionalities which were least adopted were ones that require integration with EMR systems, i.e., voice recognition for EMR and EMR management (available in one hospital). In addition, only two hospitals reported that their HIS could interface with information kiosks, electronic patient cards and mobile devices.

Figure 4. The criteria of HIS in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of level 7 can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.

Laboratory Information System

Results from our assessment show that all hospitals reported their LIS met basic criteria (Figure 5). All the LISs could manage hospital laboratory orders and results as well as generate reports. Most of these systems could automatically send orders to and receive results from laboratory machines. One hospital reported their LIS could receive results from lab instruments but the orders could only be entered manually. The national dictionary system for laboratory tests was adopted across all the hospitals to standardize lab coding. Four LISs satisfied all the Advanced criteria, which include the ability to integrate with an HIS for exchanging orders and results, set alert threshold for lab results, manage chemical supply stock, and manage lab specimens.

Figure 5. The criteria of LIS in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of 2 levels can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.

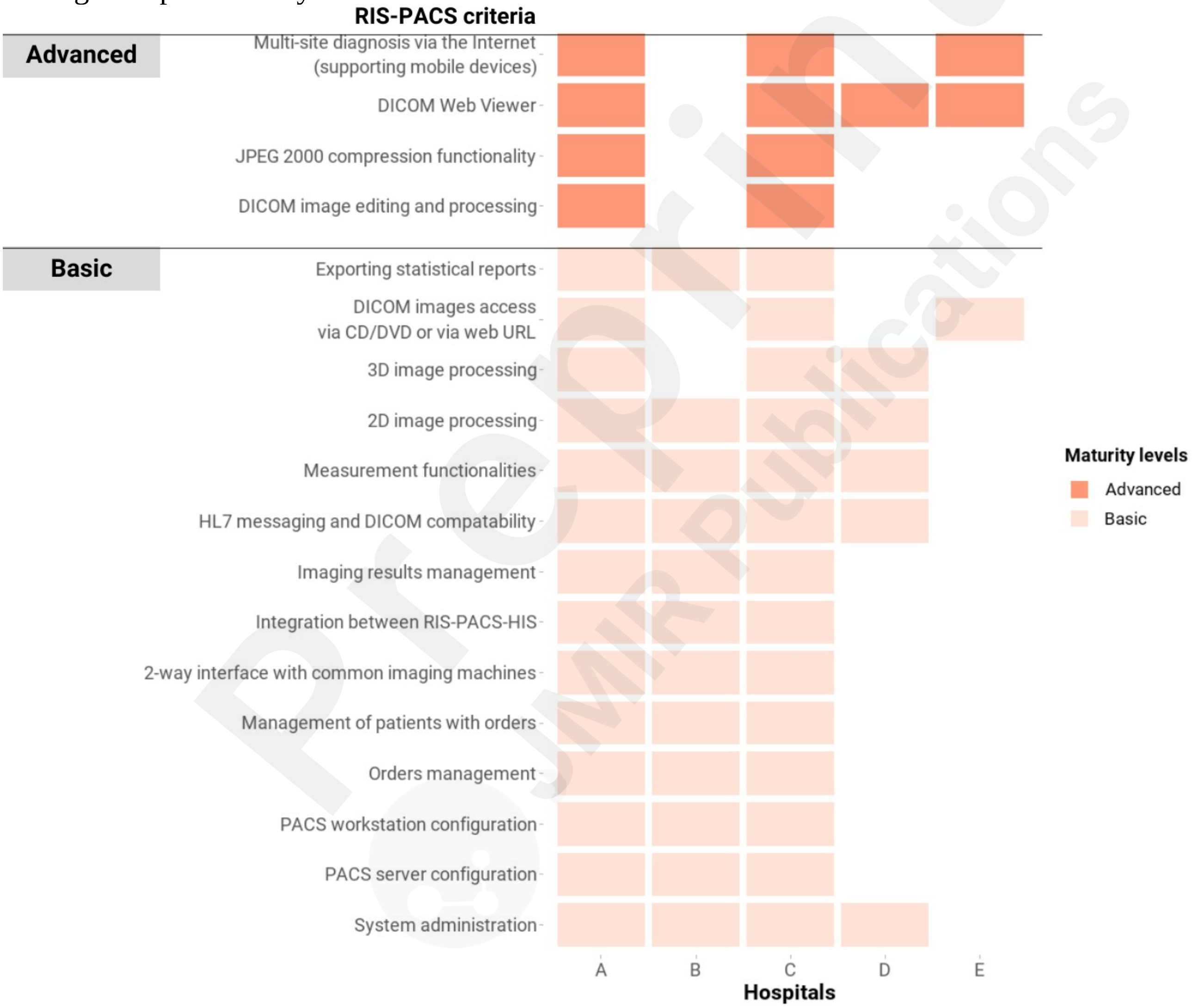
Radiology Information System - Picture Archiving and Communication System

All five hospitals were using an RIS-PACS at the time of doing this assessment although with different degrees of implementation (Figure 6). Two hospitals (A and C) had an RIS-PACS that met all the Basic and Advanced criteria. One (hospital B) nearly satisfied all the Basic Advanced criteria, while RIS-PACSs from the other two hospitals (D and E) were largely operated by external vendors and only implemented a limited number of RIS-PACS functionalities.

The RIS-PACS meeting the Basic criteria were able to retrieve Digital Imaging and Communications in Medicine (DICOM) images of common imaging modalities such as X-ray, magnetic resonance imaging, and ultrasound through 2-way interfaces. The RIS, PACS, and HIS could exchange orders and images with each other based on HL7 messaging standards. The PACS could convert DICOM images to JPEG format and any changes to images in the PACS can be promptly updated to the HIS. Order and radiologists' reading result management, measurement, and reporting were also ensured by these RIS-PACS.

Hospitals meeting the Advanced criteria of RIS-PACS could carry out multi-site consultation through web-based access to DICOM images. One hospital reported that their PACS had adopted the HL7 FHIR standard.

Figure 6. The criteria of RIS-PACS in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of 2 levels can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.

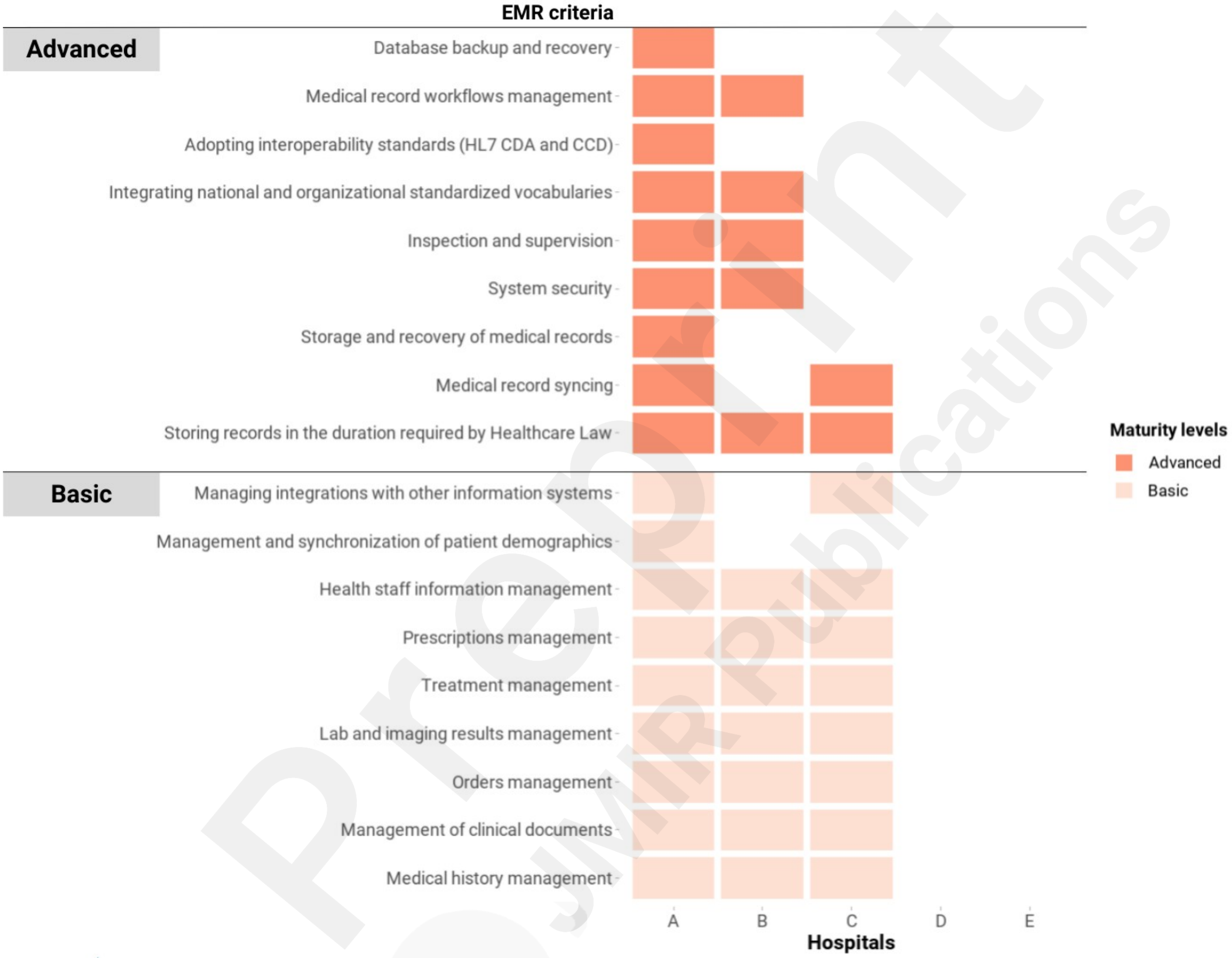


Electronic Medical Records

Generally, EMR status was less mature than other domains (Figure 7), with only three hospitals reporting that they were implementing EMR systems. Only one of these satisfied sufficient criteria in the Regulations for EMR systems[24], which allowed the hospital to completely remove paper records archiving for back-up and legal purposes. Nevertheless, the other EMR systems could provide clinical data management functions including managing medical history, clinical documents, orders, lab and imaging results, treatment, and prescription.

An EMR system fully compliant with the Regulations for EMR could ensure storage capacity, back-up, and recovery capability. Interoperability standards such as HL7 Clinical Document Architecture (CDA), HL7 Continuity of Care Document (CCD), and HL7 FHIR were implemented. Information security measures such as authentication, audit trail, and encryption were in place. Digital signatures were used to authenticate medical records and orders.

Figure 7. The criteria of EMR in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of 2 levels can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.

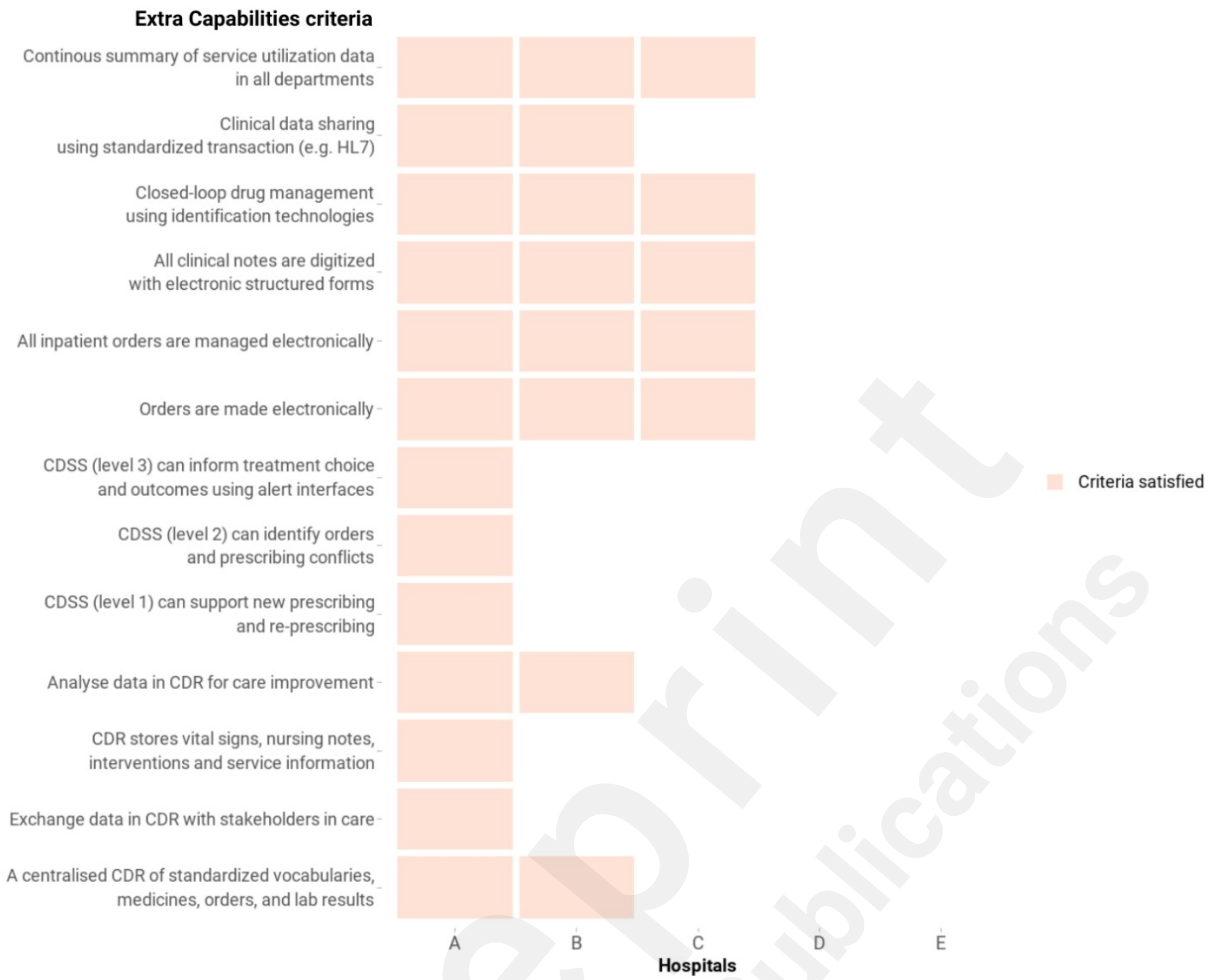


[a]The *Maturity levels* color coding only represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. A hospital needs to meet all the criteria of the lower levels to reach a higher level.

Extra capabilities

The availability of these capabilities in these hospitals is shown in (Figure 8). Notably, one hospital reported achieving all the extra capabilities with two hospitals confident about sharing clinical data using HL7 standards (although not currently sharing data at the time of the study). The adoption of CDR and CDSS to support clinical tasks such as giving orders, prescribing and clinical decision making was only available in one hospital. However, all the hospitals with an EMR system reported being able to digitize all clinical documents and allow clinicians to give orders in the electronic environment.

Figure 8. The Extra Capabilities criteria in the Vietnam HIT Maturity Model met by the hospitals in this assessment.



Discussion

Summary of key findings

This study assessed the implementation of some key digital health systems in 5 top-tier hospitals in Vietnam. Overall, many processes related to administration, laboratory, and medical imaging were digitized in the hospitals through the implementation of HIS, LIS, and RIS-PACS systems on the basis of adequate basic IT infrastructures. The maturity of these systems, however, varied significantly between the hospitals although all were large top-tier hospitals. EMR systems were implemented in 3 out of 5 hospitals. While these systems could satisfy a wide range of data collection needs, their integration with CDR and CDSS was sparse.

Comparison with prior works

This is the first original research study that examines digital health systems implementation in Vietnamese hospitals since the Vietnam MoH enacted the National EMR plan and the Vietnam HIT Maturity Model in 2018 as part of the sector-wide digital transformation agenda[11,21,24]. Academic publications on adoption and implementation of hospital-based digital health systems are limited[2]. Most of the previous studies were conducted before 2014 and focused on readiness to transition from paper-based processes to electronic systems rather than describing DHT adoption[28, 29]. Other publications described the development and pilots of technical solutions such as LIS, data retrieving from medical devices, and data visualization for EMR. We could not find any research exploring the digital systems currently used in hospitals in Vietnam[30-33].

A study by Muinga et al. showed that most of the digital health systems adopted in Kenyan public hospitals from 2014 to 2016 were aimed at administrative and billing purposes rather than supporting clinical tasks[34]. Systems such as LIS, RIS, and PACS, when available, were usually stand-alone and lacking interoperability with other systems in the same institution[34]. Our study found all the Vietnamese hospitals assessed had gone beyond administrative systems and had utilized clinical information systems that are interoperable with each other, especially with HIS. EMR, as a crucial clinical information system, had been implemented in 3 out of 5 hospitals to capture a wide range of data such as medical histories, clinical documents, orders, and test results. However, the adoption of EMR capabilities such as CDR and CDSS was limited even though these are large teaching hospitals at the top tiers of the Vietnamese health system. This is inferior to the digital maturity status of public hospitals in Turkey as over 98% of the Turkish hospitals surveyed reported having CDR regardless of their size[35]. In addition, CDSS for medication orders and non-medication orders were implemented in 71% and over 57% of Turkish public hospitals, respectively. The early use of EMRAM to guide the national digital healthcare transformation may be an important facilitator for this wide adoption of CDR and CDSS in Turkish hospitals[36].

The Vietnam HIT Maturity Model[11] that this study's questionnaire was based on and the HIMSS EMRAM[9] shares several similarities regarding the assessment criteria. For example, CDR, CDSS, closed-loop medication administration, and electronic documentation are included in both models, including the maturity stage that these systems belong to. However, the EMRAM particularly focuses on EMR capabilities, in which systems such as laboratory, radiology, and pharmacy are seen as EMR's ancillaries while CDR and CDSS are the crucial components deciding the maturity of the EMR system. All the ancillary systems need to be implemented as early as in Stage 1 of the EMRAM. The Vietnam's Model, on the other hand, splits the "ancillaries" to basic and more advanced maturity levels along the digital maturity roadmap. Systems such as CDR and CDSS are the "Extra Capabilities" that should be met alongside other domains such as HIS, LIS, RIS, and PACS so that a hospital can move up to a higher digital maturity level. While the division of digital systems into smaller milestones may allow hospitals to better benchmark their digital maturity and plan for the investments, the requirement to meet maturity criteria across multiple domains can be challenging to hospitals in a resource-limited setting like Vietnam. Duncan et al. discussed that most published maturity models, including EMRAM, were not assessed in developing countries, questioning their applicability in cultures different than developed countries such as the United States and the United Kingdom[10]. In addition, the implementation of EMRAM poses the risk of investing in complex systems without meeting the organization's local needs[37].

Integrating new DHTs into the existing system

Most hospitals participating in this assessment had a low adoption rate of EMR, CDSS, and CDR. The low number of "complete" EMR systems may be caused by the cost required to upgrade infrastructure and a lack of the required technical capabilities to meet all the requirements in the EMR regulations. It is difficult for hospitals with incomplete EMR systems and CDR to embark on using CDSS. Integrating, implementing, and maintaining CDSS is usually complex and depends on multiple factors such as the availability and interoperability of data, especially from EMR, an updated knowledge base, and understanding of the hospital's workflow[38].

The wide availability of HIS, LIS, and RIS-PACS systems in the hospitals participating in our study suggests digital systems for administrative and billing purposes and simpler systems to implement tended to be prioritized. Many hospitals may choose to implement clinical documentation functions before upgrading the other components to fully comply with the EMR Regulation. This leads to an increasing amount of medical data that will be electronically available and useful for AI training and implementation even without full EMR implementation. However, the scarcity of CDSS, CDR, and standard-based data sharing experience found in this assessment suggests challenges that stakeholders may face when seeking to integrate AI-enabled solutions in Vietnamese hospitals. While these challenges should be explored by further research, we believe a lack of national frameworks and detailed guidance for meaningful and secure clinical data exchange is one of the key barriers. In addition, a recent publication from Ho et al. mentioned exploring system interoperability, collaboration with local stakeholders from multiple disciplines, and understanding local data sharing policies as some of the key considerations for applying AI in Vietnam[39].

Strengths and limitations

We examined the adoption of digital health systems using the criteria published in the Vietnam HIT Maturity Model. This is a national benchmarking tool for HIT implementation in Vietnamese public hospitals, making it familiar to the hospitals participating in this study.
Only 5 public hospitals at the top-tier level were included in this assessment so our findings do not represent the overall picture of HIT implementation in Vietnamese hospitals. The hospitals participating in this study were ones that had previous interest in DHT implementation and digital health innovation research. However, it is likely that those at the low tiers and located in the less-funded regions have lower digital maturity across all the domains, especially in EMR and the extra digital capabilities.

Implications for further study

The adoption of digital health systems in this study was assessed on technological criteria such as the software functions that were available. Future work could address other dimensions related to the implementation of these systems such as the organizational context and human factors[40]. The use of qualitative methods can be particularly suitable for such research questions as they can unveil the characteristics unique to technology adoption in hospitals of LMIC settings like Vietnam.

Conclusion

This study showed that several major public hospitals in Vietnam have had sound digital infrastructure in place to support the fundamental administrative and clinical tasks such as patient management, insurance claim, laboratory results management, and medical imaging inspection. Most of the hospitals have implemented EMR systems to a basic level that prioritizes data collection, management, and access. Nevertheless, the more advanced level of data management and use via CDR and CDSS was not common. This can be seen as a barrier for the introduction of new DHTs in these hospitals. Along with the increased amount of data collected into the systems, the adoption of HL7, DICOM and other international standards could be seen as a facilitator for new DHTs such as AI-based CDSS to be implemented in these hospitals.

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Authors' contribution

	DMT	NTD, CMD, HNH, LMK, NPH, DTTH, HTLT, TVD	LMY	CLT	CP
Substantial contributions to the conception	Yes	Yes	Yes	Yes	Yes

OR design of the work; OR the acquisition, analysis, OR interpretation of data for the work	Yes Yes Yes	Yes Yes	Yes Yes	Yes Yes	Yes Yes
Drafting the work or revising it critically for important intellectual content	Yes	Yes	Yes	Yes	Yes
Final approval of the version to be published	Yes	Yes	Yes	Yes	Yes
Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved	Yes	Yes			
Funding acquisition				Yes	Yes

Data availability

Access to the data set generated during and/or analyzed during this study is available through OUCRU's managed data access scheme. Further details on the data request procedure can be found at https://www.oucru.org/data-sharing-policy/.

Disclosure of generative AI use

The authors attest that there was no use of generative artificial intelligence (AI) technology in the generation of text, figures, or other informational content of this manuscript.

Conflicts of Interest

None declared.

Abbreviations

AI: artificial intelligence
CCD: Continuity of Care Document
CDA: Clinical Document Architecture
CDR: clinical data repository
CDSS: clinical decision support system
DHT: digital health technology
DICOM: Digital Imaging and Communications in Medicine
EMR: electronic medical record
EMRAM: Electronic Medical Record Adoption Model
FHIR: Fast Healthcare Interoperability Resources
HIMSS: Healthcare Information and Management Systems Society
HIS: hospital information system
HIT: health information technology
HL7: Health Level Seven
LIS: laboratory information system
LMIC: low- and middle-income country
MoH: Ministry of Health
PACS: picture archiving and communication system
RIS: radiology information system

SDG: Sustainable Development Goal

Multimedia Appendix 1

The Hospital Health Information Systems Questionnaire.

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Supplementary Files

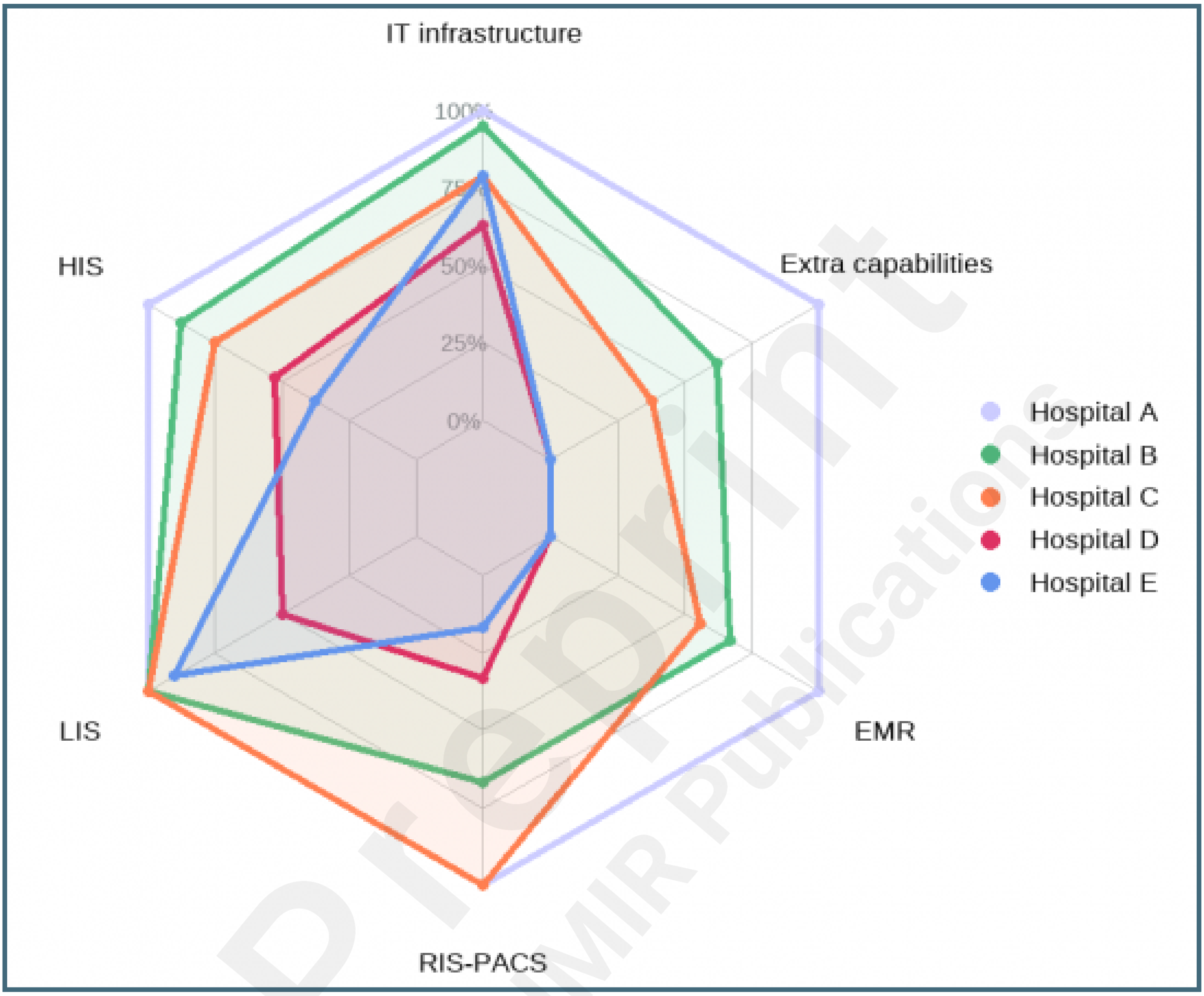
Please Select A Types

The Health Information Technology Maturity Model issued in Circular 54/2017/TT-BYT by the Vietnam Ministry of Health[11] (The header column shows the maturity levels while the header row shows the domains and extra capabilities that make up the maturity levels).

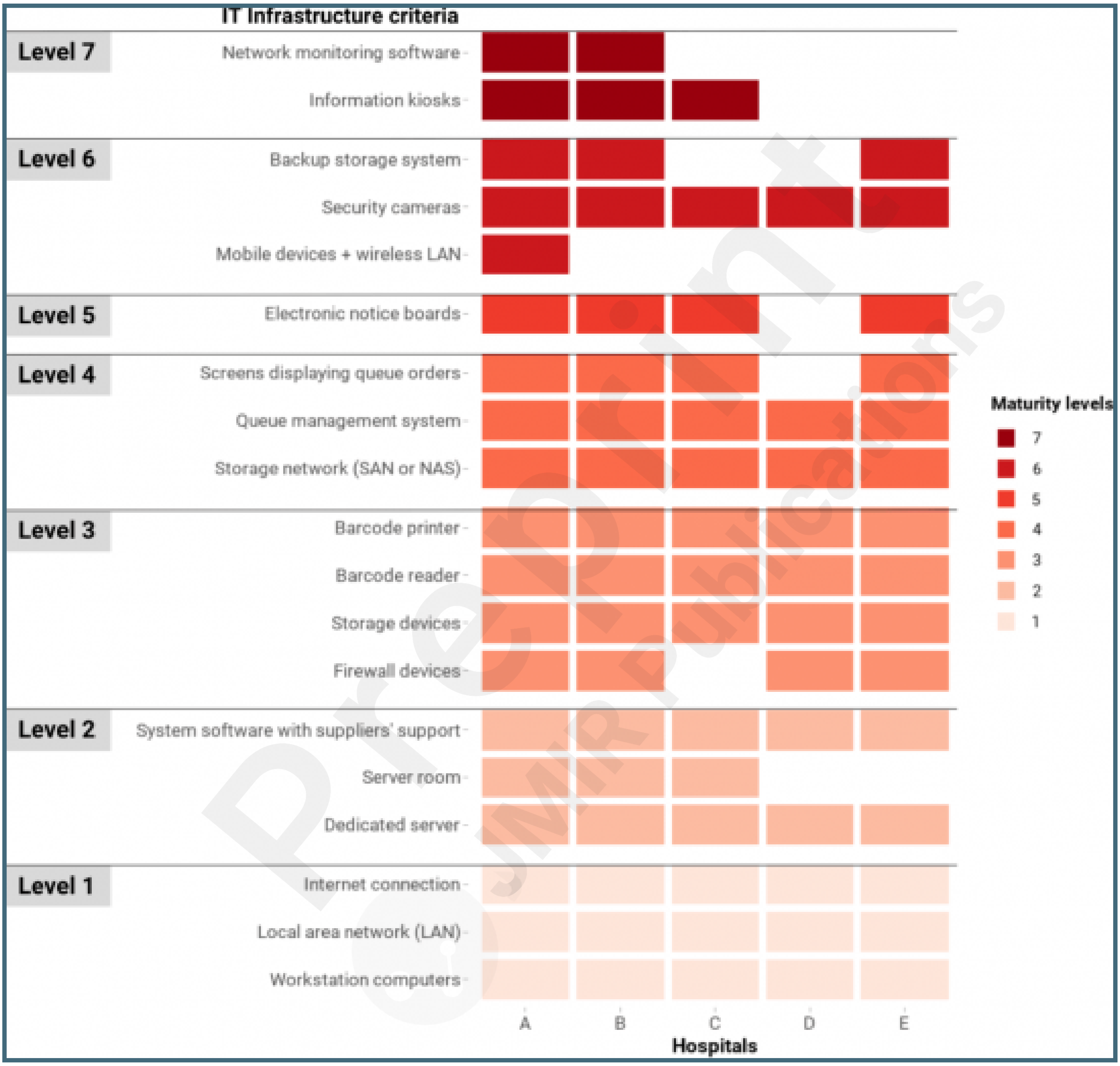
	IT infrastructure	HIS	LIS	RIS-PACS	EMR	Administration and Operation Software	Security and Information Safety	Non-functional Criteria	Extra capabilities
HIT Level 7	Level 7	Level 7	Advanced	Advanced	Advanced	Advanced	Advanced	Advanced	- "Paperless" hospital if all relevant criteria are met - CDSS level 3 supporting doctors' decisions related to treatment protocols and treatment results using suitably customized templates - Data in CDR is analysed to improve care quality, patient safety, and care efficiency - Clinical data can be readily shared for stakeholders in care coo rdination based on HL7 standards - Continuous reports of hospital services using the data collected
HIT Level 6 (Smart hospital)	Level 6	Level 6	Advanced	Advanced	Basic	Advanced	Advanced	Advanced	- CDSS level 2 providing: evidence-based warnings for treatment i.e. health and medicine advices, drug information/interaction check, and initial order/prescription violation indentification rules - All structured forms i.e. progress notes, consultation notes, problem lists, and discharge summaries are digitalized - Closed-loop management of drugs, using identification technologies to assist drug administration
HIT Level 5	Level 5	Level 5	Advanced	Advanced	n/a	Basic	Basic	Basic	PACS can replace physical films
HIT Level 4	Level 4	Level 4	Advanced	Basic	n/a	Basic	Basic	Basic	- PACS allows doctors to access images outside the imaging department - Electronic ordering - Electronic management of inpatient orders
HIT Level 3	Level 3	Level 3	Basic	n/a	n/a	Basic	Basic	Basic	- Electronic records having digital vital sign records, nursing notes, medical procedures, and surgical procedures are stored in CDR - CDSS level 1 assisting electronic prescription (new or historic prescription) - Pharmacy information available in the hospital network and supported with CDSS
HIT Level 2	Level 2	Level 2	n/a	n/a	n/a	n/a	n/a	n/a	- A CDR consisting the nomenclature and coding systems, pharmacy, orders, and test results (if available) - Data in CDR can be shared between stakeholders for care coordination
HIT Level 1	Level 1	Level 1	n/a	n/a	n/a	n/a	n/a	n/a	Patient's information can be accessed electronically

Figures

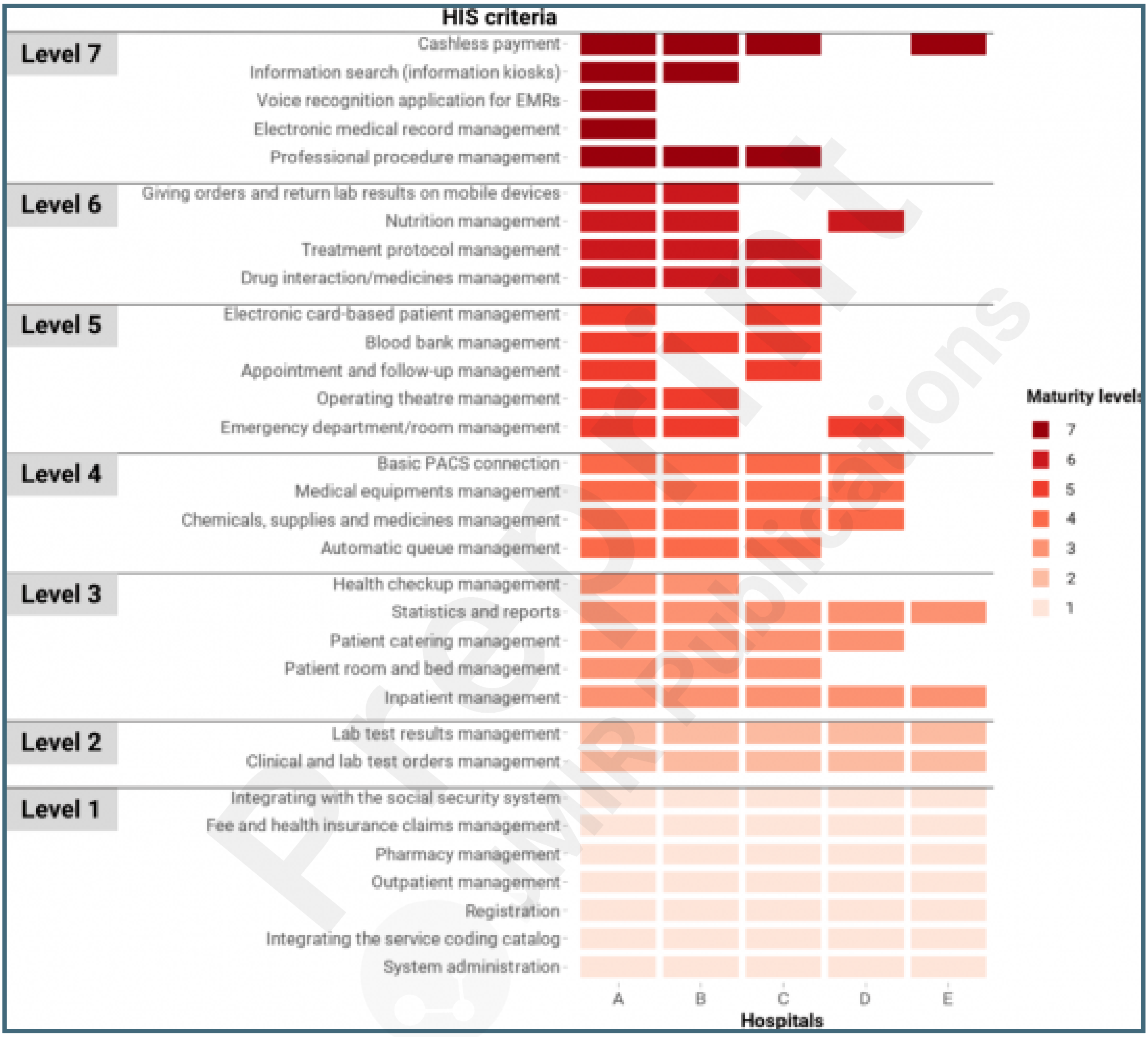
The percentage of HIT criteria satisfied in each domain across the 5 hospitals (IT infrastructure, EMR, LIS, HIS, RIS-PACS, and extra capabilities).



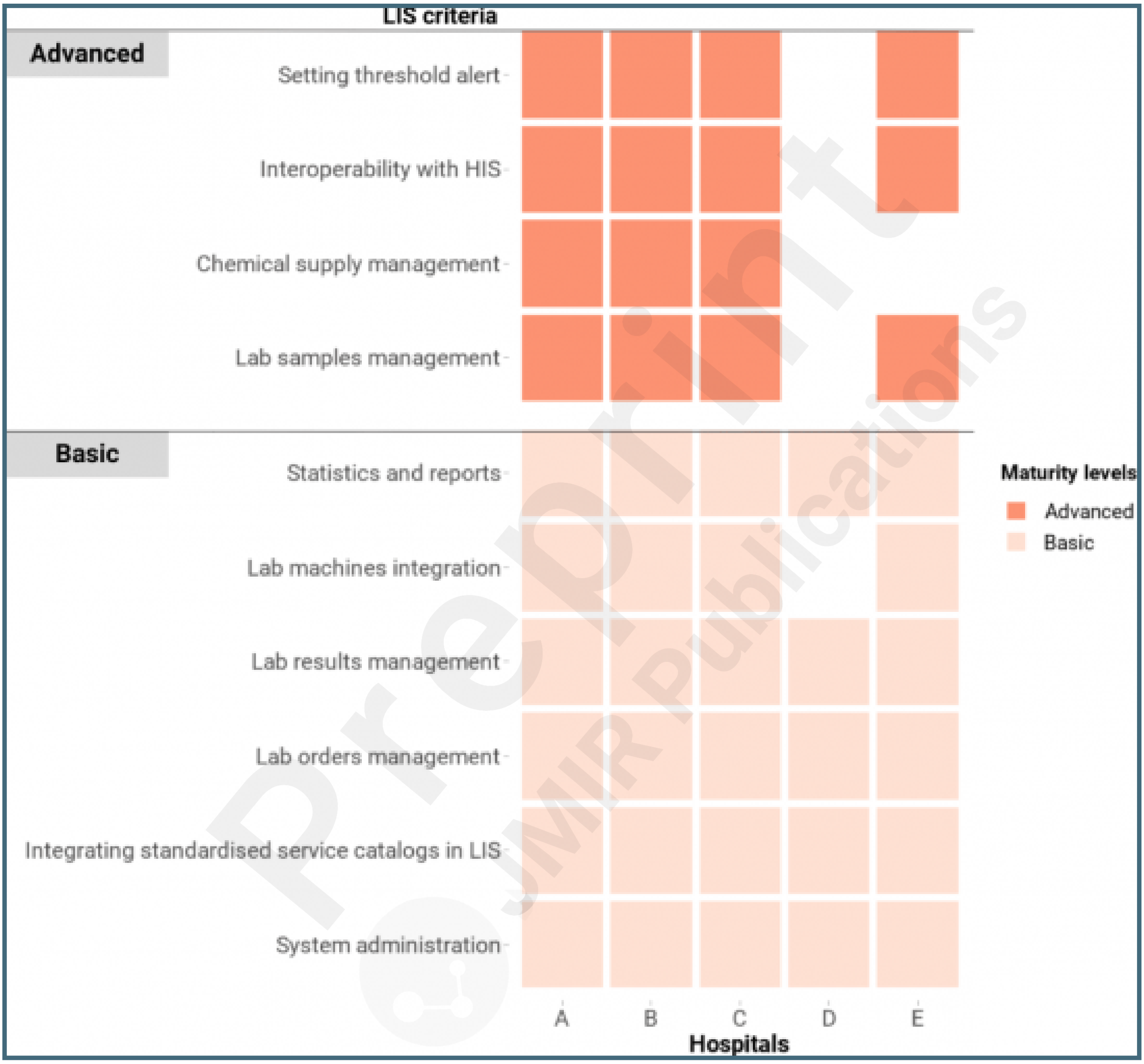
The criteria of IT infrastructure in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of level 7 can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.



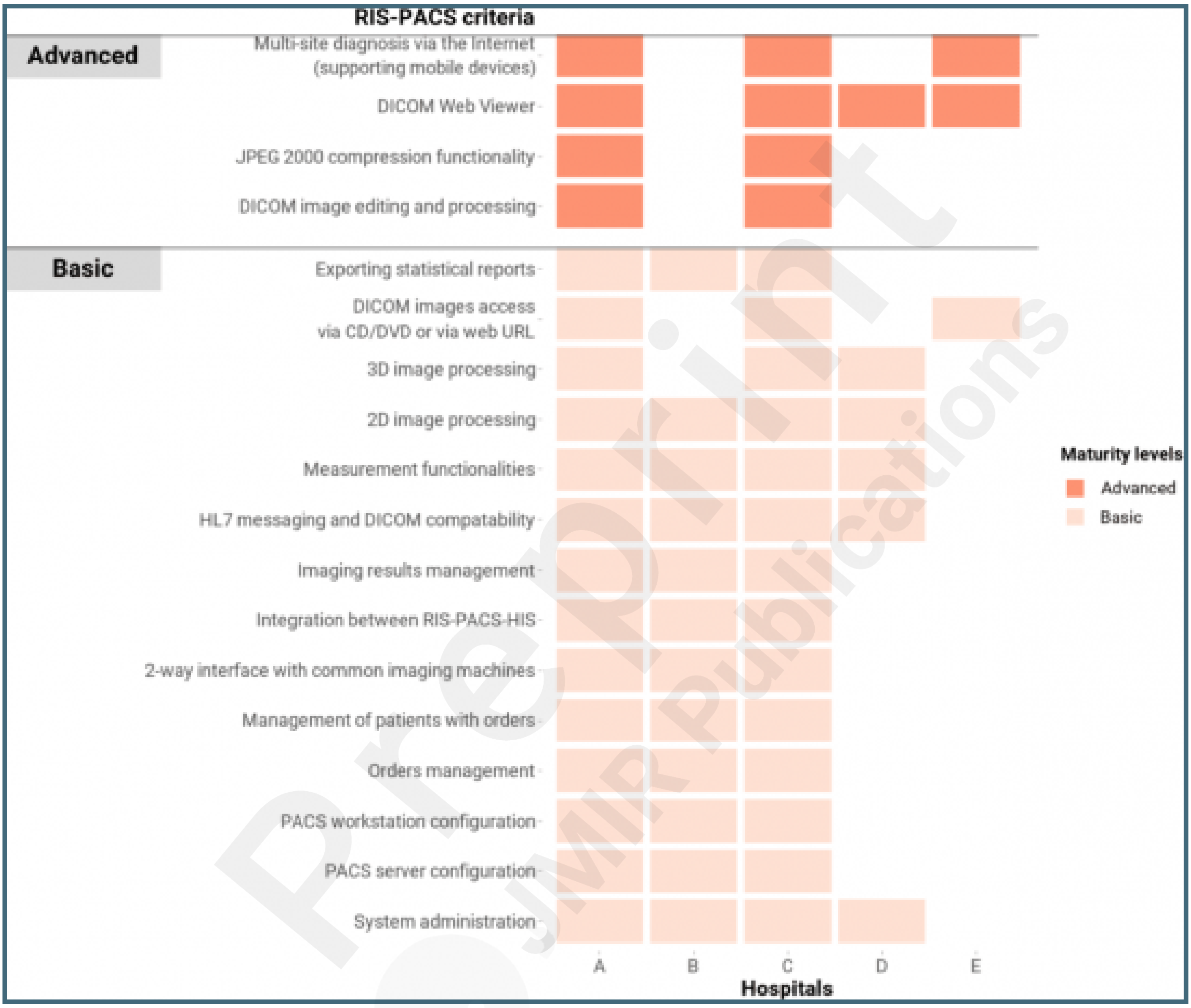
The criteria of HIS in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of level 7 can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.



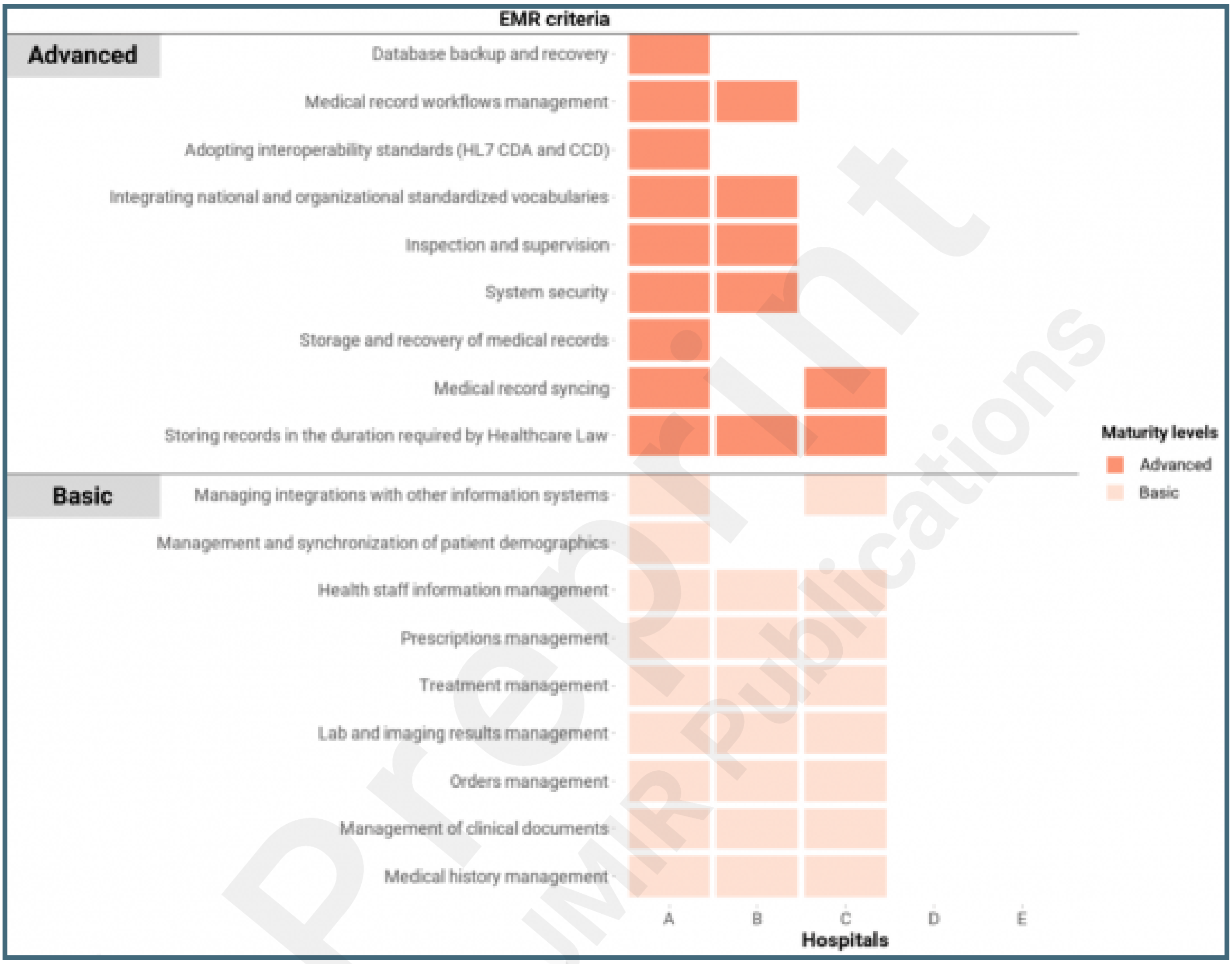
The criteria of LIS in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of 2 levels can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.



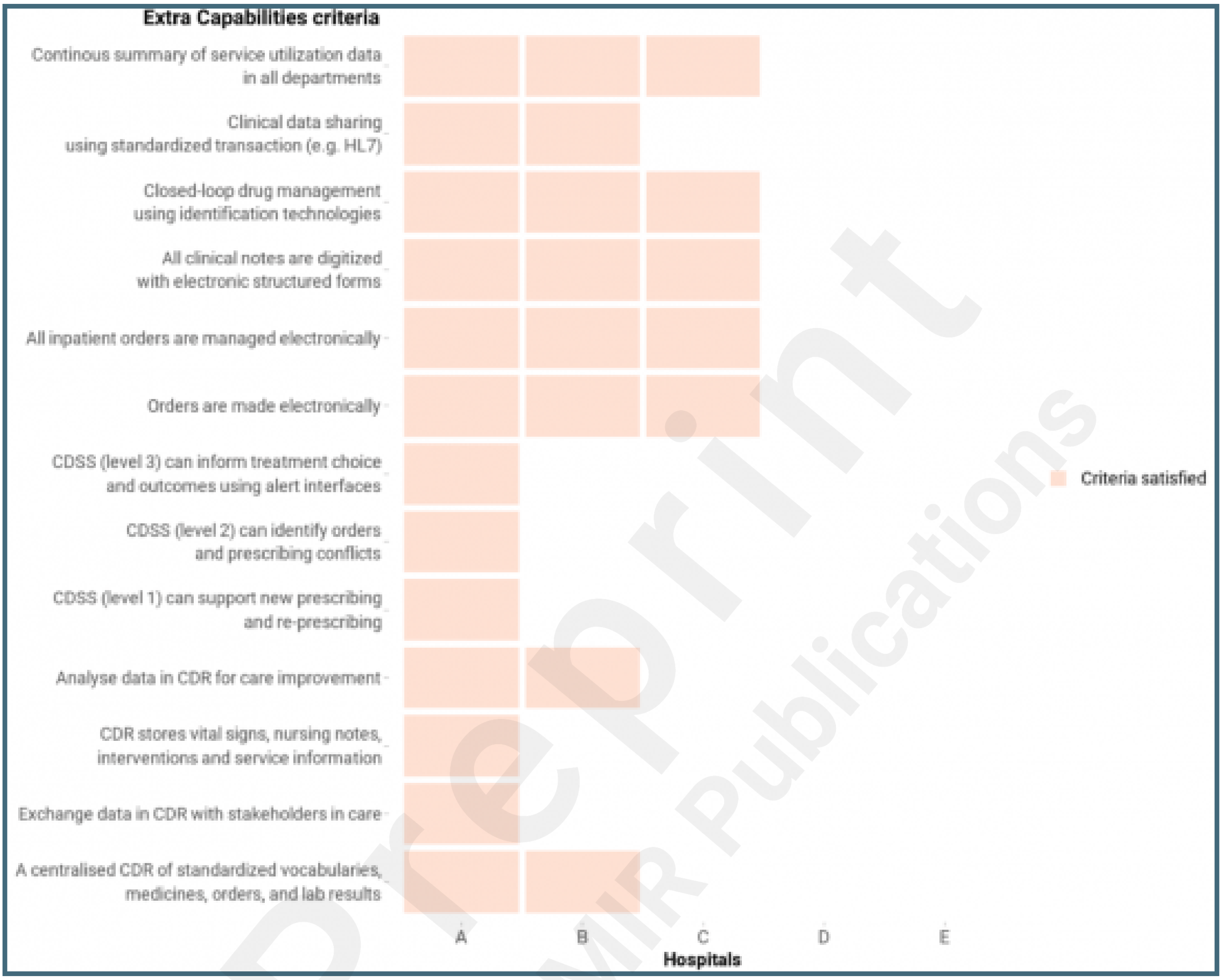
The criteria of RIS-PACS in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of 2 levels can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.



The criteria of EMR in the Vietnam HIT Maturity Model met by the hospitals in this assessment, where a maximum of 2 levels can be obtained. The color coding represents the maturity group that a criterion belongs to, not the whole domain's maturity level of a hospital. Whilst for official purposes a hospital needs to meet all the criteria of a lower level to progress to a higher level, this figure represents any criteria attained.



The Extra Capabilities criteria in the Vietnam HIT Maturity Model met by the hospitals in this assessment.



Multimedia Appendixes

The Hospital Health Information Systems Assessment Questionnaire.
URL: http://asset.jmir.pub/assets/79a6bf9a612e9f24843d2e1c64a58591.docx